CIE AS PHYSICS (9702) • QUESTION PAPER

9.2 Potential Difference & Power

Syllabus Section: 9. Electricity **Paper Type:** Structured Questions (Paper 2) **Total:** 6 Questions (68 Marks)

Question 1 Easy [9 marks]

Part (a) [2 marks]

State the **symbol** and the **SI unit** for potential difference.

Part (b) [3 marks]

Electric charge and potential difference are related.

(i) State an equation that relates electric charge and potential difference

[1]

(ii) Write one **joule** in terms of units of **potential difference** and **electric charge**

[2]

Part (c) [2 marks]

A lamp is powered by a 3.0 V cell.

Calculate the work done on the lamp by a single electron in this circuit.

Part (d) [2 marks]

Three more identical 3.0 V cells are added in series to the circuit.

> Calculate the new work done per electron.

Question 2 Easy [8 marks]

Part (a) [1 mark]

State the correct arrangement for the equation linking power, current and resistance.

Part (b) [3 marks]

A current of 3.6 A passes through a 20 Ω resistor.

Calculate the power transferred to the resistor. Give your answer to 2 significant figures.

Part (c) [3 marks]

Complete the sentences.

Choose answers from the box.

Each answer can be used once, more than once or not at all.

current	**kinetic**	**potential difference**
thermal	**gravitational**	**resistance**

When the ..... is increased, more energy is transferred to each coulomb of charge passing through the cell.

For a given potential difference, if ..... is increased, then the power transferred to a component is increased.

A high current means that more power is wasted to the ..... store of the surroundings.

Part (d) **[1 mark]**

Draw a voltmeter, connected correctly, to the circuit diagram in Fig 1.1 to measure the **combined** potential difference across the bulb and the variable resistor.

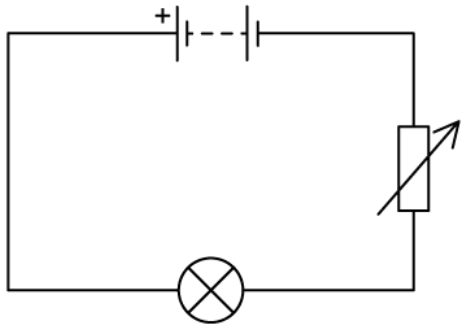


Fig 1.1

Question 3 Easy [13 marks]

Part (a) [4 marks]

The table shows a range of household appliances, supplied with 230 V, and their power ratings.

Appliance	Power rating	Current drawn
Desk lamp	14 W	
55 inch OLED TV	125 W	
Toaster	480 W	
Microwave	900 W	

Complete the table by adding the values for the current required for each appliance.

Part (b) [3 marks]

Calculate the electrical work done by using the toaster on full power for 3 minutes.

Part (c) [3 marks]

Calculate the amount of charge that flowed through the toaster circuit during the 3 minutes of use. Give your answer to 2 significant figures.

Part (d) [3 marks]

Calculate the resistance of the toaster.

Question 4 Medium [17 marks]

Part (a) [3 marks]

The power P dissipated in a resistor of resistance R is measured for a range of values of the potential difference V across it.

The results are shown in the table in Fig. 1.1.

V /	V^2 /	P /
1.00	1.0	0.214
1.71	2.9	0.58
2.25		1.011
2.67		1.43
3.00	9.0	1.80
3.27	10.7	2.176
3.50	12.3	2.43

Fig. 1.1.

For the table in Fig. 1.1

i) Add the correct units.

[1]

ii) Complete the missing data.

[1]

iii) Identify errors in the recorded the data and correct them.

[1]

.....

.....

.....

Part (b) [7 marks]

Complete the graph of the data on Fig. 1.2. below.

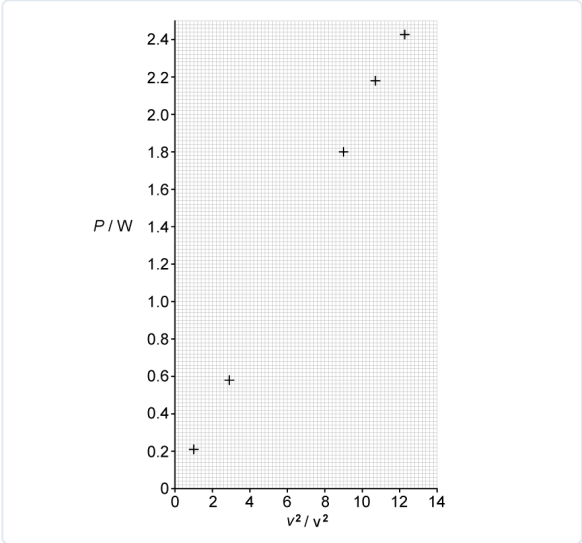


Fig. 1.2.

i) Plot the two missing points **and** complete the graph.

[3]

ii) Determine the gradient of the graph.

[3]

iii) Use the gradient of the graph to obtain a value for *R*.

[1]

Part (c) **[7 marks]**

The rest of this question is based on the data in the table Fig. 1.1 in part (a).

i) Determine the value of R when V = 3.50 V.

[1]

ii) The uncertainty in V is ± 0.01 V.

The uncertainty in P is ± 0.05 W.

Calculate the percentage uncertainty in the value of R calculated in part (i).

[3]

iii) Hence calculate the uncertainty in the value of R.

[1]

iv) State and explain whether the value of R you calculated in part (i) is consistent with the value of R you determined from the gradient in part (b).

[2]

Question 5 Medium [10 marks]

Part (a) [3 marks]

The table in Fig. 1.1 shows the power output of three heaters. All the manufacturers claim that the heaters are rated to 1500 W.

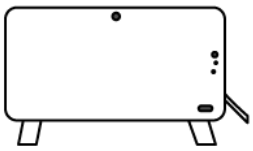
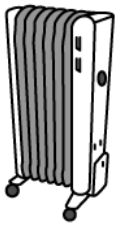
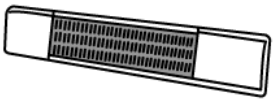
Heater	A 	B 	C 
Potential difference / V	90	220	240
Current / A	17	8.0	6.5

Fig. 1.1

Determine the most efficient heater.

Part (b) [2 marks]

Explain how these heaters can be made more efficient.

Part (c) [3 marks]

Calculate the work done by heater A when operating for 20 minutes.

Part (d) [2 marks]

Two different heaters X and Y with the same internal resistance, $5R$ use currents of I and $2I$ respectively.

State the magnitude of the power of Y in terms of I and R.

Question 6 Medium [11 marks]

Part (a) [4 marks]

A student is investigating the brightness of light bulbs in different arrangements connected to a 230 V power supply. One such arrangement is shown in Fig. 1.1.

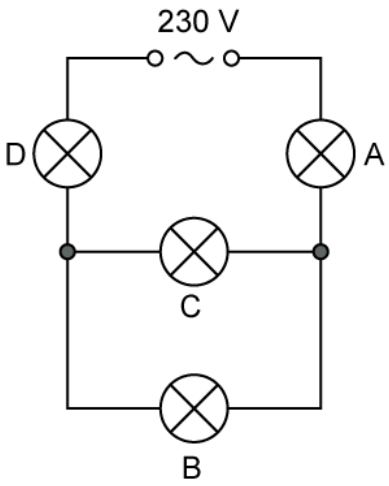


Fig. 1.1

All the lightbulbs are identical. They are all rated 15 W and 230 V. The student claims the bulbs in this arrangement are all of equal brightness.

Determine and then explain whether the student is correct.

Part (b) [3 marks]

A different experimental arrangement is shown in Fig. 1.2. Two bulbs are connected in series with a corresponding voltmeter in parallel.

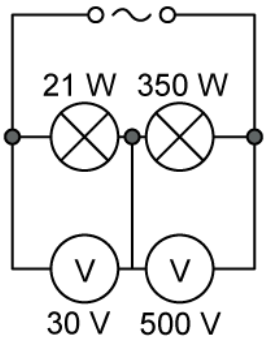


Fig. 1.2

Calculate the current needed for both lamps to operate.

Part (c) [2 marks]

Explain why the voltmeter must be connected in parallel with the bulb.

Part (d) [2 marks]

In a further experimental set-up as shown in Fig. 1.3. Four identical bulbs are all connected in parallel with each other and to a power supply of 230 V.

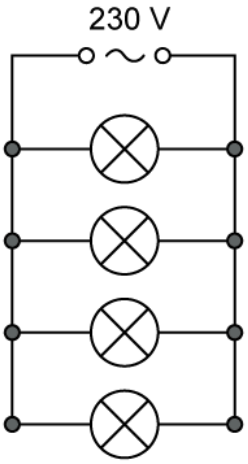


Fig. 1.3

State and explain the value of the potential difference across each bulb assuming there is no internal resistance in the power supply, the bulbs or the wires.